

Anway Pimpalkar

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EDUCATION

Johns Hopkins University , Baltimore, MD	GPA: 4.0 / 4.0	Aug 2023 – May 2025
<i>MSE, Biomedical Engineering (Focus: Neuroengineering, Medical Devices)</i>		
<i>Thesis Advisor:</i> Jeremy D. Brown, PhD (Haptics and Medical Robotics Lab)		
<i>Relevant Courses:</i> Neural Implants and Interfaces, Biomedical Instrumentation (Honors), Neural Prosthetics, Haptic Interface Design, Rehabilitation Engineering, PCB Design, Precision Care Medicine		
College of Engineering Pune , Pune, India	CGPA: 8.24 / 10.00	Aug 2019 – May 2023
<i>BTech, Electronics and Telecommunication Engineering</i>		
<i>Relevant Courses:</i> Sensors and Automation, Microcontrollers, Signals and Systems, Analog Filters, Machine Learning, Deep Learning and Edge Intelligence, Digital Signal Processing, Configurable Logic & Processor Design, CMOS VLSI Design, IoT, Embedded Systems and RTOS Design		

AWARDS AND HONORS

First Class with Distinction (Highest awarded merit), <i>College of Engineering Pune, India</i>	May 2023
Best Paper Award , <i>International Conference on Biomedical Engineering Science and Technology, India</i>	Feb 2023
Mitacs Globalink Research Award (CA \$9,915) , <i>Mitacs and Dalhousie University, Canada</i>	May 2022
Best Project Award , <i>IEEE National-Level Project Competition, India</i>	Jun 2021
Freshman Most Enthusiastic Rower Award , <i>College of Engineering Pune Boat Club, India</i>	Mar 2020

RESEARCH EXPERIENCE

Johns Hopkins University, Haptics and Medical Robotics Lab ↗	Baltimore, MD
<i>Graduate Research Assistant</i> , Post-stroke neurorehabilitation and soft haptic interfaces design	Jan 2024 – Present
<i>Advisor:</i> Jeremy D. Brown, PhD ↗	
- Exploring sensorimotor losses experienced at the fingers post-stroke, particularly in a precision grip and finger individuation context. Building a rehabilitation device capable of recording 3D isometric forces with 2mN resolution at 10kHz through neurotraining tasks. Demonstrated 25% improvement in human motor rehabilitation outcomes by integrating haptic feedback compared to visual feedback alone. <i>Work accepted for oral presentation of results at the Biomedical Engineering Society Annual Meeting 2024.</i> Abstract ↗ - Developing and testing a novel, naturalistic pneumatic haptic interface capable of simultaneous stimulation of multiple fast and slow adapting mechanoreceptors in the same location. Device can render pressures up to 3.3kPa across a vibration range of 0-145Hz. Conducting human subject experiments to validate enhanced haptic saliency, and exploring applications in prosthetics and stroke rehabilitation. <i>Device demonstrated at the IEEE Haptics Symposium 2024.</i> Video ↗ - Piloting an online learning platform to provide self-paced lectures with interactive features, enhancing student engagement through personalized feedback and on-demand instructor interaction, in the Fall 2024 offering of the Haptic Interface Design class. - Assessing wrist proprioceptive and motor capabilities in stroke patients, identifying active and passive hand synergies to inform future rehabilitation strategies.	
Harvard University, Biohybrid Organs and Neuroprosthetics Lab ↗	Cambridge, MA
<i>Research Fellow</i> , Neural stimulation for cancer metastasis reduction, urological surgical techniques	Apr 2024 – Aug 2024
<i>Advisor:</i> Shriya S. Srinivasan, PhD ↗	
- Investigated neural-cancer interactions to uncover mechanisms driving glioblastoma multiforme metastasis. Developed implantable RF-powered wireless vagus nerve stimulators delivering upto 1kHz pulses for tumor growth modulation. - Established over-the-skin electrical stimulation paradigms to modulate melanoma tumor growth in mice models, with pilot studies revealing potential correlations between stimulation frequency and tumor innervation in histological analysis. - Developed and tested a novel muscle graft-based surgical method for treating stress urinary incontinence in rat models, avoiding muscle fatigue and invasive urethral devices, offering shorter operative times and reduced technical complications compared to existing techniques.	

RESEARCH EXPERIENCE (CONTINUED)

Johns Hopkins University, Medical Ultrasound Imaging and Intervention Collaboration [↗](#)

Baltimore, MD

Graduate Researcher, Photoacoustic retinal prosthesis

Oct 2023 – Jan 2024

Advisors: Emad Bector, PhD [↗](#) and Jeeun Kang, PhD [↗](#)

- Investigated retinal stimulation using photoacoustic techniques, enabling flexible spatial targeting and deeper retinal activation. Aimed to reduce axonal stimulation and bypass limitations of the neural code.
- Contributed to the optimization of Pd-based photoacoustic-sensitive layers in PDMS, demonstrating precise piezo excitation of human stem cell-derived retinal ganglion cells at 10Hz, and recorded activations using MEA. [Abstract](#) [↗](#)

IIT Bombay, Human Motor Neurophysiology and Neuromodulation Lab [↗](#)

Mumbai, India

Research Intern, Investigating the prospective component of sense of agency in human motor control

Jan 2023 – Aug 2023

Advisor: Nivethida Thirugnanasambandam, PhD [↗](#)

- Explored the outcome-predictive capabilities of functional networks involved in sensorimotor integration and quantified how they strengthen the sense of agency. Utilized machine learning and feature engineering to identify novel pre-motor spatiotemporal EEG features, achieving up to 76.8% accuracy in predicting outcome probabilities related to the sense of agency and intentional binding.
- Revealed strong correlations between specific EEG features and motor outcome probabilities, highlighting distinct neural patterns preceding voluntary movements that contribute to accurate predictions. [Bachelor's Dissertation](#) [↗](#)

College of Engineering Pune, Departments of Electronics and Applied Sciences

Pune, India

Undergraduate Researcher, MRI processing methods to identify anatomical changes in chronic MDD

Aug 2022 – Aug 2023

Advisors: Mahesh Shindikar, PhD, Rashmika Patole, PhD and Ketaki Kamble, PhD

- Developed MRI-based skull stripping pipelines using various flavors of 2D U-Net architectures, achieving 99.75% accuracy on test data, with Dense U-Net outperforming other architectures by enhancing feature reuse across layers.
- Addressed multi-scanner variability in MRI segmentation by implementing data augmentation techniques, improving robustness and reliability of skull stripping. *Awarded Best Paper at ICBEST.* [Paper](#) [↗](#)
- Worked toward identifying regional brain volume changes and neuroimaging biomarkers in sMRIs of patients diagnosed with Major Depressive Disorder using deep learning, aiming to predict the feasibility of early risk evaluation.

Dalhousie University and Mitacs, Agricultural Mechanized Systems Group [↗](#)

Nova Scotia, Canada

Globalink Research Intern, Automating wild blueberry harvesters to reduce operator cognitive load

May 2022 – Aug 2022

Advisor: Travis Esau, PhD [↗](#) | *Funded by Mitacs Globalink Research Award*

- Developed automated mechanized systems to reduce human labor in wild blueberry harvesting, incorporating real-time point cloud segmentation and volumetric analysis with $\pm 10\%$ accuracy using Time-of-Flight and RGB imaging tools.
- Designed a control system for automated harvester head height adjustment, utilizing prescription maps generated from multi-spectral drone data of wild blueberry fields.

Queliz Lifetech Pvt. Ltd. [↗](#)

Pune, India

Research Intern, Electronics control systems for hand rehabilitation devices in burn patient recovery

Jun 2021 – Sep 2021

- Built control systems for a hand rehabilitation device to support patients with acute burns in MCP, DIP, and PIP joints.
- Developed computerized models of finger movement in flexion-extension cycles and a GUI-based feedback system to control lead-screw actuation mechanisms.

TEACHING ASSISTANTSHIPS

Principles of the Design of Biomedical Instrumentation, Primary: Nitish V. Thakor, PhD

Fall 2024

- Mentoring labs on advanced circuit design, amplifier design, and medical applications such as ECG and EMG circuits.
- Guiding student projects focused on gaming accessibility for differently-abled individuals and wearables development.

Mechatronics, Primary: Jeremy D. Brown, PhD

Spring 2024

- Built a ZigBee communication system to enable real-time location polling of autonomous air hockey robots. [GitHub](#) [↗](#)
- Led labs on sensors and actuators interfacing, mechanical design, and control systems for maze-navigating robots.

CONFERENCE PRESENTATIONS

* denotes presenter

- C1 **Pimpalkar A.***, Rai D., Bartels J.U., Xu J., Brown J.D., "Visual-haptic feedback enhances finger individuation in a virtual precision grip neurotraining task," Accepted, *Biomedical Engineering Society Annual Meeting*, Oct 2024. [Abstract ↗](#)
- C2 **Pimpalkar A.***, Ameta P.*, Dalia A.*, Brown J.D., "Pneumatactor Arrays for High Frequency Vibrotactile Feedback," *United States Senate AI Caucus: Robotics Demo Day*, Apr 2024.
- C3 Song H.* et al. [MUSiiC and Applied Physics Lab Collaboration, including **Pimpalkar A.**], "Towards visual function restoration through photoacoustic stimulation," *Association for Research in Vision and Ophthalmology Annual Meeting*, May 2024. [Abstract ↗](#)
- C4 **Pimpalkar A.***, Ameta P.*, Dalia A., Brown J.D., "Pneumatactor Arrays for High Frequency Vibrotactile Feedback," *IEEE Haptics Symposium*, Apr 2024. [Video ↗](#) [Work-in-Progress Paper ↗](#) [Poster ↗](#)
- C5 Harris C.*, **Pimpalkar A.**, Aggarwal A., Yang P., Chen X., Overby-Taylor C., Greenstein J., Stevens R.D., "Surgical risk prediction using an explainable deep learning approach applied to pre-operative 12-lead electrocardiograms," *Johns Hopkins University WSE/SOM Research Retreat*, Feb 2024. [Poster ↗](#)
- C6 **Pimpalkar A.***, Patole R., Thirugnanasambandam N., "Demonstrating the Prospective Component of Sense of Agency using Machine Learning," *COEP Electronics and Telecommunication Symposium*, May 2023. [Poster ↗](#)
- C7 **Pimpalkar A.***, Patole R., Kamble K., Shindikar M., "Performance Evaluation of Vanilla, Residual, and Dense 2D U-Net Architectures for Skull Stripping of Augmented 3D T1-weighted MRI Head Scans," *International Conference on Biomedical Engineering Science and Technology*, Feb 2023. **Best Paper Award** [Paper ↗](#)
- C8 **Pimpalkar A.***, Patole R., Kamble K., Shindikar M., "Evaluating U-Nets for Skull Stripping of Augmented T1-weighted MRI Scans," *No Garland Neuroscience*, Feb 2023. [Oral Presentation ↗](#) [Poster ↗](#)

PUBLICATIONS IN PROGRESS

- J1 **Pimpalkar A.**, Slepyan A., Thakor N.V., "Vibration sensing in prosthetic grasp is informative of object stiffness, texture, and required grip force at first contact." In preparation for *IEEE Transactions on Medical Robotics and Bionics*, 2024.
- J2 Harris C., **Pimpalkar A.**, Aggarwal A., Yang P., Chen X., Overby-Taylor C., Greenstein J., Stevens R.D., "Surgical risk prediction using an explainable deep learning approach applied to pre-operative 12-lead electrocardiograms." In preparation for *JAMA Cardiology*, 2024.
- J3 **Pimpalkar A.**, Niture D., "Towards Contactless Elevators with tinyML using Person Detection and Keyword Spotting." In review at *Smart and Sustainable Built Environment*, 2024. **Best Project Award** [Preprint ↗](#)

NOTABLE ACADEMIC PROJECTS

Vibration Sensing for Stiffness Estimation and Grasp Force Modulation in Prosthetic Hands Jan 2024 – Present
Advisor: Nitish V. Thakor [↗](#), *Honors Instrumentation (JHU)* [Video ↗](#)

- Progressing tactile sensing in prosthetics through a biomimetic sensing approach with custom multilayered sensors that mimic both fast and slow adapting human mechanoreceptors, achieving accurate stiffness estimation (MSE = 0.7 Shore A) and rapid contact detection (latency = ~1.43ms).
- Designed and integrated a PCB with a piezoresistive and piezoelectric components into the TASKA prosthetic hand, enabling real-time capture of force and vibration data for closed-loop actions on the device.

Surgical Risk Prediction with Explainable Deep Learning on Pre-Operative ECGs Aug 2023 – May 2024
Advisor: Robert Stevens, MD, Joseph Greenstein, PhD and Casey Overby-Taylor, *Precision Care Medicine (JHU)*

- Predicting major post-surgical adverse outcomes by developing a fusion deep learning model using pre-operative 12-lead ECG waveforms and clinical variables, achieving high accuracy in predicting myocardial infarction (AUC = 0.858), in-hospital mortality (AUC = 0.899), and composite outcomes (AUC = 0.835).
- Utilized counterfactual ECG generation to enhance model transparency, allowing visualization of key morphological features influencing predictions, significantly improving explainability in post-surgical risk assessment.

TinyMLElevator: Smart Elevator System using Embedded Machine Learning Jan 2021 – Jun 2021
Advisor: Deeplaxmi V. Niture [↗](#), *Microproject (COEP)* [GitHub ↗](#) **Best Project Award**

- Developed a contactless elevator system using tinyML, achieving 83.34% accuracy in person detection, 80.5% in keyword spotting, and overall system latency under 5 seconds. Built an edge-device multitenant inference engine for real-time operations, reducing computational load and implementation infrastructure costs.

SKILLS AND CERTIFICATIONS

Programming & CS	Python, MATLAB, C/C++, R, LaTeX, Lua, Verilog, machine & deep learning, data science, embedded machine learning (TinyML), web programming and design
Electronics	Analog and digital circuit design, Arduino/ESP/STM, high-speed and high-current PCB design, FPGA, RF design, soldering, mechatronic integration
Fabrication & Design	Flexible and rigid FDM 3D-Printing, SLA resin printing, silicone molding, CAD (AutoCAD, Fusion 360, Shapr3D), laser cutting, Adobe Photoshop and Illustrator, DaVinci Resolve
Medical Research	Human subject and patient studies, rodent surgical studies, experimental paradigm design, clinician-in-loop iterative design
Certifications 	TinyML Professional Certificate, <i>Harvard on edX</i> Deep Learning Specialization, <i>DeepLearning.AI on Coursera</i> Python for Research, <i>Harvard on edX</i> Neurohacking in R, Neuroscience for Neuroimaging, <i>Johns Hopkins on Coursera</i> Aeronautics and Human Spaceflight, <i>MIT on edX</i> CS50 Web Programming with Python and JavaScript, <i>Harvard on edX</i> Computer Science & Biology, <i>All India Council for Technical Education</i>

LEADERSHIP, SERVICE AND TEAMWORK

JHU Gupta-Klinsky India Institute , Hopkins-India Nexus Conference Operations Team	Present
JHU Biomedical Engineering , Master's Program Representative	Present
IEEE Haptics Symposium , Student Volunteer	2024
COEP, Pune City Sculling and Indoor Rowing Team , 1X, 2X, 4X Sculler	2020 – 2023
COEP Impressions Cultural Festival , Head of Events and Concerts (2021-22), Coordinator Organized and executed 25+ annual events with 10k attendees, coordinating a team of 150 members, managing concerts, competitions, celebrity engagements, and overseeing expenditures.	2019 – 2022
COEP Data Science and Artificial Intelligence Club , Research Lead	2020 – 2022
Canadian Society of Bioengineering Annual Meeting , Student Volunteer	2022
COEP मित्र [Marathi: friend] Mental Health & Wellness Center , Student Volunteer	2020 – 2022
YOUmanity Pune , Director of Public Relations	2015 – 2018

Website Version

Updated October 5 2024